

Computational classification of SOD2 protein in coral and symbiotic algae from public genomic and transcriptomic data

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ABSTRACT

SOD2, or Manganese/Iron Superoxide Dismutase, plays a crucial role in maintaining cellular health across the tree of life. Existing literature is vague in identifying types of SOD2 proteins expressed in both corals and their symbiotic algae; most agree on the presence of a) mitochondrial targeting MnSOD in corals, and b) likely Mn dependent SOD2 in algae. This paper presents evidence for four classes of SOD2 proteins between coral hosts and their symbiotic algae. Public genomic and transcriptomic data reveal mitochondrial targeting and secreted versions of SOD2 among a wide range of cnidaria species, and chloroplast targeting (with a bipartite leader architecture) and secreted versions of SOD2 among several symbiodinium species. The mature/catalytic region of SOD2 is well conserved within each class, but highly differentiated between classes. This paper presents the mature region HMM profiles for each class, with threshold scores high enough such that they are predictive of leader type.

Introduction

We found all K04564 matched gene model derived, HMM searched, or RNAseq derived sequences, filtered them through a set of rules that checked key residues, then looked upstream for possible M / start sites and ran the sequences through DeepLoc. For SP and mTP, we limited M to XXX upstream of PF00081. For SP_cTP, we limited M to XXX upstream of PF00081. It's possible there's a different architecture with an M that's not in these ranges, this paper does not address that.

Prev work

Old alignment work <https://pmc.ncbi.nlm.nih.gov/articles/PMC4416395/>

Mn supplementation may be good <https://www.nature.com/articles/s41598-018-34994-4>

Mn dependence <https://journals.plos.org/climate/article?id=10.1371/journal.pclm.0000897>

Results

** Four groups with clean separation, with mature sequence predictive of leader

IQ tree of mature enzyme sequence (3 AA of PF00081 to 100 AA of PF02777) forms 4 clades, with *sp_cnidaria* being paraphyletic, the rest are monophyletic, hard to assign metals from this as sequence similarity may not correspond

The mature enzyme sequence, w/o leader, form distinctive clades that do not overlap. Furthermore the HMM threshold scores for mature enzyme group profiles show a large drop with F=1 (i.e. no FPs or FNs) at very clear boundaries

Mature enzyme sequence correlates with leader, and appear to also with target compartment

Generated 202 K04564 matches from our genomic and RNase datasets, reduced down to 114 unique mature enzyme sequences that possess all key residues and spans from 3rd AA of PF00081 to 100 AA of PF02777, so removing any partial matches due to bad genome assemblies or truncated gene models.

this is all_{K04564}_matches_{mature}_{region}.faaandthe.tsv file

12 out of 144 fell below the above mature profile thresholds, rest belong to one of the 4 clades

4 are similar to or is CAE7459428.1 and is just below the threshold for *sp_ctpalgae*; CAE7459428.1 would have been a hit for *sp_ctpalgae* if its SP signal was stronger.

8/144, or 6

** SOD2 like sequences split into two distinct groups in coral hosts, targeting mitochondria and the secretory pathway, with high concordance and evolutionary lineage in the mature protein sequence Leader evidence across species from genomic and transcriptomic datasets

Mature protein sequence alignment, concordance

Mature protein sequence evolutionary lineage

Likely function and regulatory mechanism - do they differ between the two classes

Final HMM profile for each group, with threshold scores for full protein and probably mature proteins, and rules

** A secreted SOD2 may be unique to coral hosts

Paraphyletic, evolutionary lineage tree forms clusters by type: sea anemone, stony, soft.

This one may be non-essential and went through some gene loss? Or the genome assemblies are incomplete.

Transcript counts from one RNAseq experiment

Missing from acropora genomes, which explains why a. tenuis transcripts does not have such a sequence.

** SOD2 like sequence targeting chloroplast is present across many symbiotic algae and related species

SP-cTP SOD2 architecture, evidence in genomic and transcriptomic datasets

See method section for algae sequence discovery in genome, and also transcriptome datasets

Mature protein sequence alignment, concordance

Mature protein sequence evolutionary lineage

Likely function and regulatory mechanism

Final HMM profile for each group, with threshold scores for full protein and probably mature proteins, and rules

c. goreau should have had one, but in CAL1167723.1, PF00081 did not match well, if we define PF00081 from position 11 not position 3, may have been ok

Should be SP-cPT? DeepLoc reports Golgi as targets for both but do not report SP for them, SignalP reports weak SP, but sequence kind of matches BGNK01003376.1_{K04564}_a3776c6cu60OZ257276.1_{K04564}_ace37290u60

Should be SP? These are the same sequence. DeepLoc thinks they are not SP and for plastid. Mature sequence definitely matches better for *SP_a*lgaebut is below threshold. CAL1167723.1_with leader_u21_PF00081_Mevm.model.scf7180000356577.1_with leader_u21_PF00081

** Secreted SOD2 from symbiotic algae with weak evidence for expression

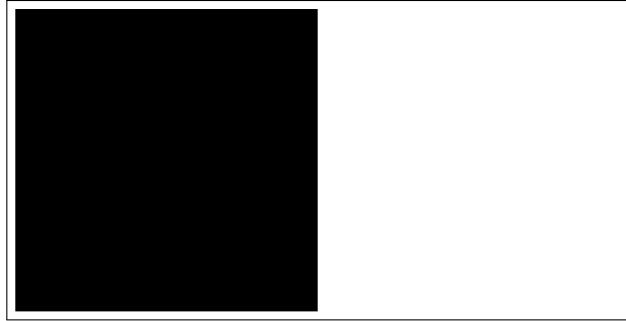


Figure 1. Sample figure caption.

Table 1. Sample table title

Part		
Name	Description	Size (μm)
Dendrite	Input terminal	~ 100
Axon	Output terminal	~ 10
Soma	Cell body	up to 10^6

Evidence for secreted SOD2 in genomic and transcriptomic datasets, including leaders. Transcript counts from one RNAseq experiment suggest very weak if any expression.

Mature protein sequence alignment, concordance

Mature protein sequence evolutionary lineage

c. goreau should have had one, but in CAL1167723.1, PF00081 did not match well, if we define PF00081 from position 11 not position 3, may have been ok

Final HMM profile for each group, with threshold scores for full protein and probably mature proteins, and rules, and how it may be confused with other classes

Methods

Rules for functional enzyme

Discussion

- Lorem ipsum dolor sit amet
- consectetur adipiscing elit.
- Aliquam dignissim blandit est, in dictum tortor gravida eget. In ac rutrum magna.

** In silico is not the same as biological validation; Lack of wet lab evidence is a weakness of this paper

Does RNA seq data present evidence for expression? Weakness of that argument. But, single time point, single stresser.

Uncertain if *sp_algae* is expressed or where it may go to.

** Unknown metal specificity

Metal specificity is a spectrum, not a binary: there are Mn-specific, Fe-specific, and cambialistic SODs that work with either ion, and the modern distribution is polyphyletic. Specificity is tuned by the redox po-

tential of the active-site metal, and several non-conserved second-sphere residues contribute to that tuning; consequently prediction methods based solely on primary sequence have not given strong predictive values. There are documented sequence/metal mismatches — e.g. the *Sinorhizobium meliloti* SOD has an iron-type sequence but preferentially incorporates Mn in vivo. The mechanistic anchor most methods lean on is the position of the solvent-coordinating glutamine (2-helix in typical FeSODs vs the C-terminal end of 2 in most MnSODs), but site-directed swaps show this is not the only determinant of metal selection. There is a model - SODa - but not super predictive. The lineage of this work: Parker Blake (1988) first showed Fe- and Mn-SODs can be distinguished by primary structure; Wintjens et al. (2004, JBC) and Wintjens, Gilis Rooman (2008, Proteins) formalized the fingerprints. A simpler, hand-usable heuristic also exists — cyanobacterial SODs can be split by a three-trait key, e.g. a signature where FeSOD's F184/A188/Q189...T280...F/Y303 is replaced in MnSOD by R184/G188/G189...G280...W303 (positions are cyanobacteria-specific).

The metal is set by the active-site environment of the protein, and localization is set by the presequence — two independent properties. Localization cannot pin the functional metal at all in the ambiguous cases.

** Localization

We checked how DeepLoc and SignalIP would call alveolata proteins that Should be mitochondria, using K00235 as control: DeepLoc detected mTP. Should be secreted, using K09490 (ER chaperone) as control: sequences from various species fall into two groups, one had SP+ER DeepLoc call, the other NLS+Cytoplasm. Stripping out the SP, DeepLoc does not by default report Plastid but reports ER and Cytoplasm as localization, and NLS, NES, and SP as signals. Should be plastid, using K02639 (ferredoxin) as control: DeepLoc detected Golgi as localization, but SP was only called for some sequences not all even though the leader sequence appears to be highly conserved. Stripping out the SP, DeepLoc reports Plastid as the localization for the truncated protein. n is very small here, as many of the sequences are essentially the same 3 protein sequences. To make this publishable we should look for different plastid targeted sequences and hope to observe the same behavior from DeepLoc.

** HMM profiles pan metagenomic or family specific

Because K04564 encapsulates the entire Fe-Mn SOD family across all domains of life (Bacteria, Archaea, and Eukaryota) [Source: KEGG Orthology K04564], a high bit-score threshold (e.g., >150) would optimize precision for eukaryotic sequences but drastically collapse the recall rate for highly divergent bacterial or archaeal orthologs. The optimization algorithm drove the threshold down to 41.53 to maintain functional annotation capacity across highly diverse phyla.

In an ancient, universally distributed enzyme like MnSOD, large stretches of the 200 amino acid sequence (such as surface-exposed loop regions and variable structural linkers) exhibit extreme sequence divergence across species. In the pHMM, these hypervariable positions have high entropy, dropping their bit-score contribution to near zero. Only the rigid structural core—specifically the N-terminal alpha-hairpin domain, the C-terminal alpha/beta domain, and the 4 metal-coordinating catalytic nodes (His-50, His-98, Asp-183, His-187)—contribute highly positive bits. A truly orthologous but highly divergent sequence will only accumulate score points at these narrow, invariant motifs, resulting in a low overall sequence bit score that still represents a true positive.

E.g. even in Alveolata, can have quite different sequences, so profiles may need different thresholds. Hard to quantify though if they are differentiating.

Using HMM profiles with rules to check for leaders and key residues. Profiles give you a fast first screen, and then the key residues make sure they are catalytically functional. Leaders verify full length proteins. Mature HMM gives you a quick way to see what class a sequence may be w/o detecting a leader which could be challenging esp in algae.

**** Considerations for RNAseq analysis**

In addition to more specific HMM profiles to match to, there is the issue of sequencing holobiont and then doing capture or removal

**** Continuing work of curating proteins**

n is small, as we get more data, the profiles get better and new profiles likely emerge. We are also interested in taking the approach to other proteins and protein families.

Gene model is difficult, needs more RNAseq data, AUGUSTUS modifications do not work. HMM of the mature protein is useful for global HMM search.